

Crypto Futures Risk Mitigation: Dynamic Rebalancing, Carry and ARP Strategies

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Abstract

This study analyzes two alternative risk premium (ARP) strategies and a near-arbitrage strategy aimed at capitalizing on large convenience yields while managing risks in cryptocurrency futures markets. Focusing on Bitcoin (BTC) and Ethereum (ETH) from March 2021 to March 2024, we test three hypotheses: (1) a long-spot, short-futures portfolio can generate positive ARPs from market inefficiencies and convenience yields; (2) high-frequency default-aversion strategies outperform unprotected counterparts by reducing forced liquidation and boosting Sharpe Ratios; and (3) protective options, such as one-touch digital options, can create near-arbitrage opportunities through volatility hedging. Through portfolio simulations, we examine methods for preserving risk premium persistence while addressing margin constraints and default-aversion. Analysis of high-frequency data suggests that these approaches mitigate major risks and yield consistently positive returns, offering insights for investors managing intermittent cash flows and long-term exposure to cryptocurrencies. The strategies also demonstrate resilience by producing returns that are uncorrelated with broader market dynamics. These findings complement broader evidence from carry-trade research, where persistent premia often emerge under limits-to-arbitrage, by showing that such mechanisms appear in amplified form within cryptocurrency futures markets.

Keywords: Cryptocurrency Futures, Bitcoin (BTC), Ethereum (ETH), Basis Trading, Alternative Risk Premium (ARP), Market Inefficiencies, Portfolio Strategies, Risk Management, Crypto Carry Puzzle, Convenience Yields, Arbitrage Strategies, One-Touch Options, Leverage, Dynamic Rebalancing, Portfolio Insurance

JEL Classifications: G11, G12, G13, G14, G15

1. INTRODUCTION

The management of fluctuations in asset prices and convenience yields is critical for investors seeking to optimize returns in cryptocurrencies and those engaged in hedging through futures or forward contracts. These fluctuations introduce risks associated with intermittent cash flows from hedging and margining activities, posing notable challenges (ESRB 2023; Czech et al. 2021). This paper explores strategic portfolio concepts aimed at mitigating such risks.

We focus on the anomaly of significant convenience yields in cryptocurrency futures markets, often referred to as the “Crypto Carry Puzzle.” In traditional asset classes, a carry puzzle relates to a strategy that earns a return mainly from “carry” (the yield or interest-rate advantage earned from holding an asset or position) delivers consistently positive average returns that seem too large or too reliable to be explained by standard risk compensation. In theory, this is a ‘puzzle’ as competition and prices should adjust so that these returns are either small or come with clearly visible risk. As for the crypto carry puzzle investors can often earn relatively high returns from crypto “carry” in ways that appear persistent. We would expect convenience yields to adjust so that these easy-to-harvest returns largely disappear or are fully offset by expected losses elsewhere, hence the puzzle arising. Our research examines whether these yields provide sufficient compensation for the inherent risks of crypto basis trading. Convenience yields in this market can reach as high as 40%, with an average around 10% (Schmeling et al. 2023). Our alternative risk premium (ARP) strategy involves going long in underlying cryptocurrencies while simultaneously shorting corresponding futures contracts. While Schmeling, Schrimpf, and Todorov (2023) document the magnitude and time-variation of crypto futures carry at the market level, their analysis does not examine how an investor might capture these premia under realistic margin, liquidation, and funding constraints. Our contribution extends this literature by developing and empirically evaluating implementable long-spot/short-futures strategies, including high-frequency default-aversion overlays and protective option structures. By translating the structural forces identified by Schmeling et al. into portfolio-level experiments, we demonstrate how these unusually high convenience yields can be harvested while mitigating the liquidation and margin risks that limit arbitrage in practice.

This basis-trade approach parallels traditional carry trades by emphasizing the cost of carry and capturing convenience yields. Unlike typical analyses focusing on high-yield versus low-yield currencies, our framework bears similarities to equity market dividend yields and FX carry strategies (Zhao et al. 2018; Cong et al. 2023; Fan et al. 2023). While convenience yields in cryptocurrency futures appear unusually large, it is important to situate this phenomenon within the broader carry-trade literature. In foreign exchange markets, systematic excess returns to currency carry portfolios are well explained by exposure to global risk factors rather than arbitrage (Lustig, Roussanov & Verdelhan 2011). Persistent deviations from covered interest parity (CIP) further illustrate how balance-sheet constraints and regulatory frictions can sustain forward–spot mispricing even in highly liquid markets (Du, Tepper & Verdelhan 2018). Commodity futures exhibit similar mechanisms: convenience yields rise when inventories tighten, producing predictable basis premia (Gorton, Hayashi & Rouwenhorst 2013). These parallels demonstrate that persistent carry premia are not unique to cryptoassets. What sets cryptocurrency markets apart is the amplification of these forces through custody frictions, fragmented liquidity, and exchange-level liquidation dynamics, contributing to the exceptionally large convenience yields documented in recent work (Schmeling, Schrimpf & Todorov 2023).

To explore these opportunities, we develop three portfolio strategies targeting unexplained convenience yields: an unprotected ARP (default-unprotected) portfolio, a

protected ARP (default-protected) portfolio, and an arbitrage portfolio. The key research questions are: (a) Can a strategy combining long spot and short futures positions yield a positive alternative risk premium? (b) Does implementing high-frequency (5-minute) default-aversion rebalancing improve performance over unprotected strategies? It is important to note that the default-aversion is defined as the investors' dislike of the possibility of the borrower to default, even when compensated by high expected returns. It leads investors to require an additional risk premium (higher credit spreads) or to limit holdings of defaultable claims even when their expected returns match those of safer assets. (c) Can including a protective call option (e.g., a one-touch digital option) alongside long spot and short futures positions create near-arbitrage conditions free of market risk?

First, we construct a portfolio with long spot and short futures positions in Bitcoin (BTC) and Ethereum (ETH), filtering for a minimum Sharpe ratio of 1.0. We simulate these portfolios to test for the persistence of an alternative risk premium. Second, we incorporate margin constraints and default-aversion into the ARP strategy, following methodologies by Alexander et al. (2021). Third, we develop an arbitrage portfolio via a one-touch digital option that can hedge against volatility drawdowns while preserving positive returns.

Our data collection relies on five-minute intervals for spot, futures, and implied volatility indices from 2021 through 2024, capturing a realistic high-frequency environment. We evaluate the attractiveness of the returns using Sharpe ratios (Da et al. 2023). Our analysis indicates that high convenience yields and market inefficiencies create strategic opportunities to earn alternative risk premiums with limited correlation to broader market behavior. This supports the needs of risk-averse investors. Additionally, our findings generate avenues for future research into advanced risk management techniques and novel derivatives products.

This paper is structured as follows: Section two outlines the background and motivation for our hypotheses. Section three details the methodology, including portfolio design and implementation. Section four presents the results, while section five discusses these findings, concludes the study, and offers directions for future research.

2. BACKGROUND AND HYPOTHESES DEVELOPMENT

Cryptocurrency markets blend characteristics of foreign exchange (high volatility, 24/7 trading) and commodities (low physical storage costs, but significant custody and security challenges). Regulatory uncertainties, fragmented exchanges, and technical complexities, including blockchain vulnerabilities, also shape these markets (Chen et al. 2023).

Despite their rapid evolution, cryptocurrency markets exhibit notable inefficiencies, similar to other asset classes. Cross-exchange arbitrage opportunities have been documented (Makarov and Schoar 2020; Shynkevich 2021), although convergence has increased since 2018 (Shynkevich 2021). Similar patterns exist in FX, where deviations from covered interest parity rose after the 2008 financial crisis (Baba and Packer 2009; Graeb et al. 2017; Tola et al. 2020). In commodity markets, physical and margin constraints often limit arbitrage, yet transient price anomalies can be exploited (Shleifer and Vishny 1997; Gromb and Vayanos 2002; Girard and Reid 2010; Sikorski et al. 2018).

2.1. ARBITRAGE OPPORTUNITIES IN CRYPTOCURRENCY FUTURES MARKETS

A key framework in futures pricing is the cost of carry, which posits that a futures price adjusts to cover holding costs over time. Thus, it is the net cost of owning and holding

an asset until a future date. It includes factors such as interest that could have been earned instead (financing cost) and storage/insurance, minus any benefit that is gained from holding the asset (like convenience or income). In simple terms: it is “what it costs (or pays) to keep the asset over time.” Deviations from this relationship signal potential arbitrage when spot and futures prices diverge. Formally,

$$F = S \times e^{(r+s-c) \times t} \quad (\text{eq.1})$$

where F is the forward price, S is the spot price, r is the risk-free interest rate, s is storage cost, c is the convenience yield, and t is the fraction of a year until delivery. The relationship highlights today's price S and is adjusted for the net cost of holding it until time t , which include the interest given up (r) plus storage/insurance (s) minus any benefit from physically having the asset (c).

Covered interest parity (CIP) further explains futures pricing in FX markets, asserting that forward–spot differentials reflect interest rate spreads:

$$\frac{F}{S} = \frac{1 + r_d}{1 + r_f} \quad (\text{eq.2})$$

where r_d and r_f are domestic and foreign interest rates, respectively. The CIP ascertains that the forward exchange rate must adjust so that investing in the domestic currency at rate r_d gives the same risk-free return as converting to the foreign currency, investing at r_f , and locking in the future exchange rate. For this reason, the forward-to-spot ratio simply matches the interest-rate difference between the two currencies. Deviations from CIP in crypto markets reflect inefficiencies that can be harnessed for arbitrage (Baba and Packer 2009; Graeb et al. 2017; Tola et al. 2020).

2.2. CONVENIENCE YIELD IN CRYPTOCURRENCIES

Convenience yield is the benefit of holding an asset, motivating investors to prefer futures over spot (Shleifer and Vishny 1997; Gromb and Vayanos 2002). In cryptocurrency markets, this yield corresponds to risks and costs of holding digital assets, which affect pricing and deter arbitrage. Although in commodities, limited storage or transport fosters the yield, in crypto it stems from custodial risks and market fragmentation. At expiration of contract, the spot and futures prices converge:

$$F_{t,T} = S_t \times e^{(r-c) \times (T-t)} \quad (\text{eq.3})$$

meaning that capturing high convenience yields should, under no constraints, yield a positive payoff upon convergence. This relationship ascertains that the futures price is today's spot price adjusted for time, where the interest (r) drives the price up, while the convenience yield (c), the benefit of holding the asset now, pulls the price down. Thus, if the convenience yield is high, the futures price tends to be relatively low compared with spot, and as they converge at expiration, that difference can turn into a positive payoff.

Prior studies on convenience yields in cryptocurrencies (Arkorful et al. 2023) highlight shifts amid short-term trading surges. Market behavior can reflect positive or negative feedback mechanisms (Karaa et al. 2021; King and Koutmos 2021; Doan et al. 2022; Ngene and Wang 2024), while futures prices often exhibit biases relative to spot (Lee et al. 2020). Comparable mechanisms have been documented in commodity markets, where convenience yields rise when inventories are scarce and basis premia contain predictive

information about risk compensation (Gorton, Hayashi & Rouwenhorst 2013), highlighting that crypto convenience yields may reflect an analogous set of frictions despite the absence of physical storage constraints.

2.3. ALTERNATIVE STRATEGIES IN CRYPTOCURRENCY MARKETS—BASIS TRADING

Arbitrage, derivatives, and basis-trade techniques are especially relevant to managing crypto market inefficiencies and volatility. Basis trading exploits price discrepancies between spot and futures. Bitcoin (BTC) and Ethereum (ETH) futures have been tradable since December 2017, allowing traders to sell futures and buy spot when a positive basis emerges. As with commodity markets, convenience yields and CIP deviations influence these setups (Franz and Valentin 2020).

Although basis trading in cryptocurrencies has unique microstructural features, it is instructive to situate the “crypto carry puzzle” within the broader literature on carry trades across asset classes. In foreign exchange (FX) markets, for example, returns to currency carry portfolios are well understood as compensation for exposures to global risk factors rather than pure arbitrage, as demonstrated by Lustig, Roussanov, and Verdelhan (2011). Moreover, persistent deviations from covered interest parity since the Global Financial Crisis show how regulatory constraints and balance-sheet frictions can prevent arbitrageurs from eliminating forward–spot differentials even in deep and liquid markets (Du, Tepper & Verdelhan 2018). Related mechanisms appear in commodity futures: when inventories are scarce, convenience yields rise and basis premia become strongly time-varying, as documented by Gorton, Hayashi, and Rouwenhorst (2013).

These parallels indicate that persistent basis premia are not unique to cryptocurrencies. However, the magnitude and variability of crypto convenience yields—reaching levels as high as 60% per annum and fluctuating sharply over time (Schmeling, Schrimpf & Todorov 2023)—suggest that crypto markets amplify these familiar limits-to-arbitrage forces due to fragmented exchanges, custody frictions, and automatic liquidation mechanisms. Cross-venue studies similarly show that structural segmentation and settlement delays contribute to recurrent price deviations across exchanges (Makarov & Schoar 2020). Taken together, these cross-asset insights provide theoretical grounding for why large crypto convenience yields may persist despite growing institutional participation.

Within this broader context, we now turn to our hypotheses, which examine whether such elevated convenience yields can be systematically harvested through long-spot/short-futures strategies (H1), whether high-frequency default-aversion mechanisms enhance their performance (H2), and whether protective option overlays can convert these premia into near-arbitrage outcomes (H3).

Hypothesis 1 (Portfolio Strategy Efficacy). A strategy combining long spot and short futures positions will earn a positive ARP, driven by high convenience yields and inefficiencies.

ARP strategies (Reid and Van Der Zwan 2019; Naya and Tuchschnid 2018; Hamdan et al. 2016) emphasize robust risk management, given crypto’s volatility. Alexander et al. (2021) discuss reducing default probability in crypto basis trades using structured protective measures. Building on high-frequency monitoring for risk mitigation, we propose:

Hypothesis 2 (Impact of Risk Mitigation Techniques). Default-aversion strategies using high-frequency rebalancing outperform an unprotected approach, lowering forced liquidations and improving Sharpe ratios.

The significant variance in basis values, as highlighted by Schmeling et al. (2023), suggests potential for effective risk mitigation. Protective options, such as one-touch calls, may further generate near-arbitrage by hedging downside risks while preserving returns (Alexander et al. 2021). Thus:

Hypothesis 3 (Role of Protective Options). Including protective one-touch options in a long-spot, short-futures portfolio can create near-arbitrage opportunities by insulating positions from market volatility.

Finally, while prior research (Alexander et al. 2021) heavily features perpetual futures, our study focuses on dated (quarterly) futures, enabling observations of price convergence across discrete contract intervals. We also employ a dynamic risk management method akin to CPPI-based default-aversion techniques, diverging from static approaches in prior literature.

3. METHODOLOGY AND RESEARCH DESIGN

We explore whether and how to build profitable portfolios by capitalizing on large positive convenience yields. Our design simulates three distinct strategies: (1) Unprotected ARP portfolio, (2) Protected ARP portfolio, and (3) Arbitrage portfolio. Each strategy is evaluated in five-minute increments. The protected ARP model adds CPPI-style dynamic adjustments to mitigate forced liquidations, whereas the arbitrage strategy includes a one-touch digital option as insurance against default.

To validate the foundational condition for these strategies—the presence of positive convenience yields—we confirm their existence in cryptocurrency markets up to March 2024, extending the findings of Schmeling et al. (2023), detailed in Appendix A.

3.1. PORTFOLIO DESIGN

We measure risk-adjusted performance via the Sharpe ratio:

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \quad (\text{eq.4})$$

where R_p is portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of excess portfolio returns. In general, the Sharpe Ratio depicts the excess return of a security for the extra risk taken. In a forward-looking perspective, we estimate expected returns and volatility, while R_f comes from fixed-income market observations.

In basis trading, returns are primarily driven by changes in convenience yields until contract expiry, with default risk stemming from margin constraints. Following Alexander et al. (2021), we consider four ways to handle potential default: (1) Do nothing, (2) Purchase explicit insurance, (3) Replicate insurance, (4) Statistically minimize default probability. The do-nothing approach portfolio establishes its positions at the beginning of the strategy and maintains them unchanged throughout the portfolio's duration. Without any limitation on position size and risk-reward requirements, this portfolio faces a significant risk

of default. Additionally, when convenience yields are low, the potential for returns is limited, leading to a Sharpe ratio that may not justify investment in the strategy.

Crypto basis trading resembles currency hedging, requiring frequent parameter updates (Gardner and Stone 1995; Froot and Perold 1997). Our portfolios comprise trade tickets for each strategic position. Unlevered portfolios assume fully funded spot and short-futures positions, whereas leveraged portfolios earn interest on residual cash. Although coin-margined (COIN-M) futures could reduce the need for holding stablecoins, here we maintain USDT-based futures data.

3.1.1 Portfolio Construction

This strategy employs no leverage - both spot and futures positions are fully funded. At inception, the portfolio must meet a minimum expected Sharpe ratio of 1.0. New positions are not initiated within six days prior to expiration, due to the reasons previously discussed. The portfolio is closed if at least one of the following conditions hold: if the convenience yield is less than or equal to zero, or if the upper margin threshold is reached, or upon contract expiration. For this analysis, the upper margin threshold is defined as a maintenance margin of 5.0%, which is a function of leverage and absolute position size. The position size range is set between USD 375,000 and USD 2,000,000 to reflect a real-world trading environment.

The five-minute closing spot and futures prices are sourced from Binance, while the risk-free rate is determined by interpolating daily USD 1-month, 3-month, and 6-month LIBOR data. The ex-ante portfolio Sharpe ratio is calculated using observable values for the convenience yield and the risk-free return. The volatility of total portfolio returns, which forms the denominator, is the difference between spot and futures returns, which is estimated by taking the highest hourly Deribit Implied Volatility Index (DVOL) value from Deribit - a proxy for expected market volatility, and estimating the variance of the difference of two correlated variables, Y_1 and Y_2 representing spot and futures value, which is given by:

$$\text{var}(Y_1 - Y_2) = V_1 + V_2 - 2r\sqrt{V_1}\sqrt{V_2} \quad (\text{eq.5})$$

where V_1 and V_2 are the volatilities of variables Y_1 and Y_2 respectively, and the parameter r is the correlation of spot and futures that has been estimated in Schmeling et al. (2023) and confirmed in this paper to exceed 0.9 for the majority of observations. The higher the positive correlation between the variables, the smaller the volatility difference between them. In essence, the measure shows the basis (spread) risk to allow size hedges and risk limits.

3.1.2 Implementation

We distinguish portfolio construction (before trading) from portfolio implementation (post-trade). Once trades are executed, we check conditions every 5 minutes to decide if a position should be closed, focusing on Sharpe ratio targets (>1.0), margin triggers, positive convenience yields, or expiry thresholds. End-of-day revaluation at 23:55 (or upon closing) provides ex-post performance metrics.

3.2. THE PROTECTED ALTERNATIVE RISK PREMIUM PORTFOLIO

Most futures-based investing aims to use minimal collateral, but leverage heightens default risk if maintenance margins are breached. We apply a Markov decision framework to rebalance portfolio weights based on prior states and incoming data. By replicating an

“insurance effect,” we dynamically reduce exposure without re-establishing it, avoiding complexity from a separate reinvestment function. This approach aligns with delta-hedging principles but remains simpler, as we do not dynamically add positions.

In our unprotected ARP sample, no portfolios defaulted because there was no leverage (5% margin required a 95% adverse price move). Here, we introduce leverage (up to 5 $\times$) and note some portfolios then default, triggering the protective strategy.

3.2.1 Portfolio Construction

Like before, each portfolio takes a long position on the spot and a short on futures, but initial margin is lower. Market-neutral exposure is maintained at a notional level. Leverage does not alter the expected Sharpe ratio (Goetzmann et al. 2002), so the same opportunity set from the unlevered case applies. Table 1 illustrates the portfolio constituents across various levels of leverage.

**Table 1: Example portfolio constituents of Portfolio #1 in BTC 09/21 Futures contract
Price Futures: 51,196.90 / Price Spot: 50,727.04**

Leverage	1.0	2.0	3.0	4.0	5.0
Portfolio Statistics					
E(return)	8.99%	11.98%	13.47%	14.37%	14.97%
E(risk)	8.98%	11.97%	13.46%	14.36%	14.96%
E(Sharpe ratio)	1.00102998	1.00102998	1.00102998	1.00102998	1.00102998
Constituents					
Long Spot	0.00098112	0.00131018	0.00147509	0.00157415	0.00164025
Short Futures	0.00098112	0.00131018	0.00147509	0.00157415	0.00164025
Initial Cash Flows					
Spot	49.77	33.54	25.17	20.15	16.80
Futures margin	50.23	66.46	74.83	79.85	83.20
Initial Forced Liquidation Futures Price					
Spot	99,834	74,236	65,703	61,436	58,876

We monitor probability of default (PoD) every 5 minutes. If PoD exceeds a maximum threshold (PoD_{max}), we rebalance down to a target threshold (PoD_{target}). In practice, these values would be set to match investor risk tolerance.

3.2.2 Implementation

First, we run all portfolios at 5 $\times$ leverage to see which default. Those that default are flagged for subsequent performance analysis. A smaller subset of (defaulted) portfolios is rerun under the insurance replication approach, applying rebalancing steps at 5-minute intervals. PoD is estimated via Monte Carlo, setting PoD_{max} = 25% and PoD_{target} = 5%. This reduces margin calls but may incur extra transaction costs (not modeled here).

3.3. THE ARBITRAGE PORTFOLIO

Arbitrage implies immediate, risk-free profit with zero initial outlay. However, in crypto basis trading, some assumptions must be relaxed. This section constructs a near-

arbitrage portfolio by combining a 5× leveraged spot–futures position with a one-touch digital call option to cover losses in a default scenario.

3.3.1 Portfolio Construction

We use a long spot and short futures position at 5× leverage plus a binary (digital) call option that pays out upon default-triggering price moves. The option expires with the futures contract. No rebalancing is needed; it is a one-time purchase of “insurance.” Only BTC and ETH contracts with an ex-ante Sharpe ratio >1.0 are considered.

3.3.2 Implementation

As in the unprotected scenario, portfolios are built, and then the option’s premium is priced by Monte Carlo (1,000 simulations). We search for the option cost that neutralizes default losses—if that cost still permits a net positive return (across default and non-default cases), we have achieved a form of “insured arbitrage.” Effectively, if the insurance premium is low enough that the strategy still has a positive net payoff after covering default losses, then the investor has an “arbitrage” that is effectively insured.

3.3.3 Transaction Costs

To assess practical feasibility, we apply trading costs to each one of our three portfolios. In this way, we are able to determine the robustness of the profitability of each of the portfolios. Our transaction cost estimates for Taker Fees were 0.10% of transaction volume for spot, 0.05% of transaction volume for futures and 12.5% of option premium for options.

3.3.3.1 The Unprotected Portfolio Transaction Costs

The transactions costs were applied on the opening of the portfolio by entering a Spot and Forward transaction, and on the closing of the portfolio if expired, defaulted or the convenience yield was less than zero. This results in all portfolios total opening and closing costs being 0.15% respectively.

3.3.3.2 The Protected Portfolio Transaction Costs

The transaction costs for the protected portfolio were applied in the same manner as for the unprotected portfolio. This resulted in all portfolios total opening costs amounting to 0.15%, BTC portfolios total rebalancing costs, given 1.6x rebalancing on average equalled 0.09%, ETH portfolios total rebalancing costs, given 3.7x rebalancing on average equalled 0.185%, and all portfolios total closing costs amounting to 0.15%.

3.3.3.3 The Arbitrage Portfolio Transaction Costs

The transaction costs were applied to the opening of the portfolio by entering a Spot and Forward transaction and buying a protective one-touch call option, and the closing transaction costs were implemented on the portfolio if expired, exercised or the convenience yield was less than zero. This resulted in all portfolios total opening costs amounting to 0.15%, BTC portfolios option transaction costs equalling 0.11%, ETH portfolios option transaction costs equalling 0.06% and all portfolios total opening costs amounting to 0.15%.

3. RESULTS

Initial Findings on Convenience Yields In verifying the presence of convenience yields in cryptocurrency markets—often referred to as the "crypto carry"—we confirmed a highly volatile yield, averaging 10%, consistent with Schmeling et al. (2023). Our study found the convenience yield was 7.23% for BTC and 6.63% for ETH, with high intra-day fluctuations. Moreover, a positive correlation between one-week and one-month changes in spot prices and convenience yields was observed, with regression slopes of 0.21 (BTC, one month), 0.30 (BTC, one week), 0.11 (ETH, one month), and 0.22 (ETH, one week) as laid out in Appendix A.

4.1. THE UNPROTECTED ALTERNATIVE RISK PREMIUM PORTFOLIO

4.1 The Unprotected Alternative Risk Premium Portfolio

4.1.1 Evidence of Alternative Risk Premium

The ARP portfolios aimed to deliver excess returns above the risk-free rate to compensate investors for the risks associated with the strategy. The Sharpe ratio is an appropriate metric.

In Tables 2 and 3 below, we summarize the results for 30,358 portfolios with a Sharpe ratio exceeding 1.0 across BTC and ETH futures spanning from June 2021 to March 2024. The analysis indicates substantial compensation for risk, consistent with ex-ante estimates, with portfolios achieving 111% of the estimated Sharpe ratios. The variation in the portfolio opportunity set is significant and readily observable. While profitable trading opportunities were available in 2021 and 2024, the landscape was markedly different in 2022 and up to the third quarter of 2023, with only limited opportunities emerging.

Table 2: Summary statistics of 16,254 portfolio analyses (BTC)

BTC	2021			2022				2023				2024
	06	09	12	03	06	09	12	03	06	09	12	03
Portfolio Selection based on Sharpe Ratio Estimate												
Expected S.R ≥ 1	8473	821	1137	0	0	0	36	6	0	0	626	5155
% of full Dataset	31.2	3.1	4.2				0.1				1.7	10.3
Sharpe Ratio Analysis												
Average Expected	1.30	1.25	1.27				1.21	1.07			1.62	2.00
Average Realised	1.42	0.98	3.81				3.15	0.52			0.62	3.18
Median Realised	1.26	1.04	3.42				1.53	0.46			0.64	1.84
% Realised	97	83	269				127	43			39	92
Portfolio Runtime Analysis (days)												
Average Expected	77	15	16				7	7			8	17
Average Realised	20	2	8				2	0			8	13
Transaction Cost Adjusted Returns												
Average Realised	1.35	0.76	0.71				-6.54	-1.01			-0.34	1.84

Tabel 3: Summary statistics of 14,104 portfolio analyses (ETH)

ETH	2021			2022				2023				2024
	06	09	12	03	06	09	12	03	06	09	12	03
Portfolio Selection based on Sharpe Ratio Estimate												
Expected S.R ≥ 1	6606	1685	280	0	3	0	0	32	0	0	387	5111
% of full Dataset	24.3	6.4	1.0					0.1			1.1	10.2
Sharpe Ratio Analysis												
Average Expected	1.35	1.25	1.11		1.26			1.13			1.37	1.99
Average Realised	1.49	0.84	27.32		1.04			0.49			0.96	1.88
Median Realised	1.25	0.93	4.54		1.04			0.45			0.92	2.02
% Realised	92	74	409		82			40			67	102
Portfolio Runtime Analysis (days)												
Average Expected	76	15	11		27			7			8	17
Average Realised	23	3	5		9			2			5	16
Transaction Cost Adjusted Returns												
Average Realised	1.42	0.41	-14.05		0.69			-0.84			-0.82	1.16

4.1.2 Transaction Cost Adjusted Results

Considering the effect of the transaction costs on all three portfolios, we see that the results are robust for most scenarios. As expected, the results of the Sharpe Ratios decrease for all, nonetheless remaining positive, with the exception of the unprotected portfolio. The unprotected portfolios suffer from transaction costs where the opportunity set (i.e. number of portfolios) is small and absolute excess return and volatility of returns are small (as seen in Exhibit 2 and Exhibit 3).

4.1.2 Risk of Default

Futures positions necessitate the deposit of collateral to ensure that the position holder does not default on any losses incurred. If the minimum maintenance margin requirement is not satisfied, the position is subject to liquidation.

This analysis assumes no leverage, requiring a 100% collateral deposit at the portfolio's inception, with the minimum maintenance margin set at 5%. Consequently, the futures price would need to increase by 95% for the portfolio to default. As a result, none of the portfolios experienced default. The shorter-than-expected portfolio runtimes, averaging 8.2 days compared to the estimated 22.0 days, significantly reduce the probability of default. Table 4 provides a brief simulation example, illustrating the probabilities of default across varying collateral levels and different runtimes, assuming a volatility of 80% per annum. As anticipated, the absence of leverage effectively reduces the likelihood of portfolio default to near zero.

Table 4: Simulated default probabilities (Monte-Carlo 1000 samples) Volatility: 80%

Days	22	8
Collateral		
100%	0.2%	0.0%
50%	5.4%	0.1%
33%	14.7%	2.1%
25%	23.4%	6.5%
20%	34.5%	13.2%

4.1.3 Variation in efficiency

The analysis reveals significant variations in the Sharpe ratio, which can be attributed to the considerably reduced portfolio runtimes in certain instances. While the return is determined by the magnitude of the convenience yield that can be captured, the shorter portfolio runtimes, as shown in Tables 4.1.1 and 4.1.2, result in very high efficiency measures.

4.1.4 Robustness

When comparing BTC and ETH portfolios, it is evident that the datasets are highly comparable across nearly all dimensions. The opportunity sets for both BTC and ETH exhibit significant variation and closely follow the spot prices of the assets, as illustrated in Figures 5 and 6. Similarly, the expected Sharpe ratios and their variations show substantial overlap throughout the dataset, as do the portfolio runtimes and the realized Sharpe ratio percentages.

In summary, out of a total of 348,765 five-minute intervals (from the start of trading to six days before expiry) in the futures series tested, 16,254 intervals for BTC and 14,104 for ETH exhibit a Sharpe ratio greater than 1.0. This represents approximately 4.7% of all intervals tested for BTC and 4.0% for ETH. None of the portfolios defaulted, as none utilized more than 95% of the full collateral amount, thereby maintaining the 5% maintenance margin requirement. Estimating the Sharpe ratio using hourly DVOL Index series proves beneficial, with the median realized portfolio Sharpe ratio across all portfolios being 1.52, compared to an overall average estimate of 1.37. An alternative risk premium is evident in the dataset; however, it is highly time-varying and tends to cluster around periods of higher average spot prices in both BTC and ETH.

4.2 The Protected Alternative Risk Premium Portfolio

4.2.1 Overview of the effects of leverage and applying portfolio protection

Exhibit 7 presents an overview of the results for two scenarios: increasing leverage to 5X and subsequently applying the protection mechanism discussed earlier. The risks associated with leverage and the advantages of reducing exposure to prevent default become apparent quickly.

Reducing the collateral from 100% to 20% (5X leverage) at the portfolio's inception significantly increases the risk of default, as the collateral is disproportionately reduced, leading to a lower futures price that triggers portfolio liquidation. Of the initial 30,358 portfolios, 5,539 (18.2%) defaulted under the 5X leverage strategy, suffering notable declines in returns both in absolute terms and relative to the broader dataset.

For a random sample of 200 portfolios, portfolio protection was applied. This strategy involved hourly monitoring and estimating the probability of default (PoD) over the next two hours. If necessary, exposure was reduced to align with the designated target PoD level, resulting in a significant positive impact on these portfolios. The parameters used for this analysis were as follows: a maximum probability of default (PoD,max) of 25%, a target probability of default (PoD,target) of 5%, with observations conducted hourly. The analysis employed 1,000 Monte Carlo iterations and a binary search accuracy of 1.625%. On average, the returns of the unlevered portfolios were fully retained with comparable Sharpe ratios. Notably, the protected portfolios had a lifespan 50-60% longer than their unprotected counterparts. This outcome is attributed to the pre-emptive reduction of exposure when the risk of default increased sufficiently to necessitate rebalancing.

In addition, the protected portfolio is more robust to the implementation of transaction cost. In the last column of Table 7, it can be seen that most portfolios experience a positive transaction-adjusted return.

Table 7: Summary statistics of 1X, 5XU (defaulted) and 5XP (protected) portfolios

	All Portfolios			BTC			ETH		
	1X	5XU	5XP	1X	5XU	5XP	1X	5XU	5XP
Portfolio Selection based on Sharpe Ratio Estimate									
Expected S.R. ≥ 1	30358	5539	400	16254	1104	200	14104	4435	200
% of full Dataset	100.0	18.2	1.3	53.5	3.6	0.7	46.5	14.6	0.7
Sharpe Ratio Analysis									
Average Expected	1.55	1.31	1.32	1.53	1.16	1.14	1.56	1.34	1.31
Average Realised	2.05	-0.80	1.11	2.11	-1.54	0.70	1.99	-0.61	1.51
Median Realised	1.35	-0.85	1.49	1.53	-1.33	1.22	1.29	-0.69	1.61
% Realised	87		121	100		107	82		123
Portfolio Runtime Analysis (days)									
Average Expected	38	66	71	39	62	61	38	71	71
Average Realised	14	8	13	14	8	14	15	9	13
Transaction Cost Adjusted Returns									
Average Realised	1.89	-0.99	1.03	1.94	-1.76	0.66	1.82	-0.84	1.42

4.2.2 Rebalancing

Table 8 details the effects of rebalancing on BTC and ETH portfolios, with REBAL indicating rebalancing events and PoD representing the probability of default. Despite setting the PoD threshold at 25% within a 2-hour interval, defaults persisted in 20% of portfolios. However, the average PoD triggering rebalancing occurred at 34%, which is higher than anticipated. This discrepancy may result from significant movements in futures prices, an underestimation of volatility from DVOL input parameters, or inaccuracies in Monte Carlo outputs based on 1,000 simulations.

Similarly, the PoD after rebalancing slightly exceeded the 5% target, with portfolios sometimes exiting rebalancing events with a PoD of up to 13%, nearly halfway to the next rebalancing trigger. Further examination of DVOL inputs and Monte Carlo simulations could

help identify the root causes of these discrepancies. A binary search algorithm with a step size of 6 was used to find the transaction notional matching the 5% PoD target, expected to provide an accuracy of +/- 1.625% concerning the transaction notional.

Most portfolios underwent rebalancing at least once: 160 for BTC and 190 for ETH in our sample, with BTC averaging 1.6 events and ETH 3.7, typically post an approximately seven-day runtime. This is expected, as the average runtime of defaulting portfolios from the unprotected cohort ranges between 8 and 9 days. Thus, these portfolios rebalanced shortly before defaulting. Notably, there was a reduction in portfolio exposure, measured as the size of the futures position at portfolio closing compared to the initial position. Position sizes were reduced by approximately 58%, with BTC at a lower reduction of 45% and ETH at a higher reduction of 68%. This can be attributed to the PoD target parameter setting, which is relatively risk-averse at 5%. These parameters were set arbitrarily to allow for various scenarios and have not been optimized. It is worth noting that significant reductions in position sizes may lead to higher transaction costs, which were not considered in this analysis.

Table 8: Effects of rebalancing

	ALL	BTC	ETH
# Portfolios	400	200	200
Defaults Statistics			
# Defaulted	78	42	36
Defaulted (%)	20	21	18
Before 1 st REBAL	50	40	10
Rebalancing Statistics			
# REBALs on Average	2.7	1.6	3.7
Average first REBAL at day	6.9	7.0	6.8
Minimum first REBAL at day	1	3	1
Effects on Futures Exposure (Portfolio closing vs Inception)			
Average reduction (%)	58	45	68
Minimum reduction (%)	25	25	34
Maximum reduction (%)	100	67	100
Probability of Default Statistics (Monte Carlo)			
Average PoD triggering REBAL	34	34	34
Maximum PoD triggering REBAL	59	56	59
Minimum PoD triggering REBAL	25	25	25
Average PoD after REBAL	6	6	6
Maximum PoD after REBAL	13	12	13
Minimum PoD after REBAL	1	1	1

Figures 9 and 10 illustrate a sample rebalancing event for a portfolio of each cryptocurrency. In the BTC case, the portfolio experiences three rebalancing events in less than two days, occurring in quick succession. In contrast, the ETH example demonstrates a longer time span with more frequent rebalancing events. Both cases effectively demonstrate how

rising prices trigger rebalancing events, which in turn increase the liquidation price of the futures position by reducing exposure and simultaneously lowering the probability of default.

In summary, increasing leverage from 1X to 5X results in 6.8% (1,104 out of 16,254) of BTC portfolios defaulting and 31.4% (4,435 out of 14,104) of ETH portfolios defaulting. Notably, the defaulting portfolios have a significantly longer average expected runtime: 62 days compared to 22 days for BTC, and 71 days compared to 20 days for ETH. This is because these portfolios were established early in the futures contract's lifecycle. The higher number of defaulting ETH portfolios is attributed to ETH's relative outperformance during the active period of the portfolios. The impact on returns for the defaulting portfolios is substantial, with the Sharpe ratio declining from a median of 1.35 at 1X leverage to -0.85 at 5X leverage on the futures position. The time to default is relatively short, averaging 8 days.

For a random sample of 200 portfolios each for BTC and ETH, the portfolios were enhanced with the previously discussed protection mechanism. The results are notable, as the portfolios were able to return to their pre-leverage levels in terms of median Sharpe ratio: 1.49 across all 400 portfolios, with 1.22 for BTC and 1.61 for ETH.

4.3. THE ARBITRAGE PORTFOLIO

4.3.1 Some Evidence of Arbitrage

Focusing on June 2021 BTC/ETH contracts at 5× leverage, we identify a set of ex-ante Sharpe ratios above 1.0. Our approach deviates from traditional arbitrage by incorporating a one-touch call option, aiming to achieve positive returns without the immediate constraints of no upfront investment or instant execution. Two primary scenarios emerge when utilizing the one-touch call option, demonstrated here with BTC: In the non-default scenario, the expected return of 3.97% is reduced by the option cost of -0.86%, yielding an expected profit of 3.11%. Conversely, in the default scenario, the portfolio achieves a neutral return (0.00%), comprised of a -3.72% loss from the base strategy, the option cost of -0.86%, and a compensating payout of 4.58% from the one-touch option. By accounting for the default rate, the insured arbitrage portfolio consistently generates a moderate but positive profit. Summary results in Table 11 illustrate positive outcomes for both BTC and ETH futures contracts. There are significant challenges to capturing this return, including, but not limited to, transaction costs; the risk that option prices deviate significantly from their theoretical value (TV); the risk of increased bid/ask spreads at lower levels of option delta; and the risk of slippage when defaults occur due to potentially high market volatility. For an example of the complexities involved in what appears to be an obvious arbitrage opportunity in early fixed income trading, see Kane (1980).

Indeed, the results of the positive return remain robust even when adjusting for transaction costs. The expected average return, the non-default return and the default rate adjusted total arbitrage return expectedly decrease once transaction costs are accounted for, nonetheless they all remain positive indicating a robustness of the results.

Table 11: Results of Arbitrage portfolios for BTC and ETH 06/21

	BTC	ETH
# Portfolios	8473	6606
Default rate %	8.47	60.46
Arbitrage Statistics		
Expected Average Runtime	77	76

Expected Average Return %	3.97	4.64
Loss given default %	-3.72	-1.41
Required portfolio protection %	4.58	1.98
Average option price in % of notional	18.83	20.04
Cost of portfolio protection %	-0.86	-0.51
Non-default case return %	3.11	4.13
Default rate adjusted total arbitrage return %	2.84	1.63
Transaction Cost-Adjusted Results		
Expected Average Return %	3.56%	4.27%
Non-default case return %	2.70%	3.76%
Default rate adjusted total arbitrage return %	2.47%	1.49%

4.3.2 Option Price Trajectory

Figures 12 and 13 illustrate the evolution of option prices over the portfolio inception dates. They show that ETH options are generally more expensive than comparable BTC options, primarily due to higher volatility. Additionally, as the options approach their expiry dates, their value tends to decrease, a phenomenon known as Theta, which represents the decay in value over time. The impact of higher expected volatility on option prices is also evident around the midpoint of the futures contract's lifetime. This change in option prices, resulting from fluctuations in the underlying asset's volatility, is referred to as Vega.

5. CONCLUSION

This paper verifies that cryptocurrency markets exhibit substantial and persistent convenience yields (the “crypto carry”) between April 2021 and March 2024, aligning with earlier studies (Schmeling et al. 2023). Elevated yields can be captured via ARP strategies, producing returns less correlated with traditional markets, making them attractive to risk-averse investors seeking diversification.

Leverage amplifies returns but also intensifies default risk, necessitating robust risk management. Our protected ARP portfolios demonstrate that frequent monitoring and dynamic rebalancing significantly mitigate forced liquidations while preserving positive Sharpe ratios. Additionally, option-based protection—specifically one-touch digital options—can approximate an arbitrage condition by neutralizing losses in default scenarios, although high implied volatility and illiquid option markets are limiting factors.

Whereas Schmeling et al. (2023) characterize crypto carry as a market level anomaly driven by volatile convenience yields and constrained arbitrage capital, our study advances the literature by evaluating concrete implementation strategies that explicitly manage these constraints. Through protected ARP portfolios, high frequency default aversion rebalancing, and option based overlays, we show how the risks highlighted in prior work can be operationally mitigated, transforming descriptive insights into practical, risk controlled portfolio designs.

Dynamic rebalancing strategies, built on real-time data, show promise for maintaining stable risk-adjusted returns. Periodic probability-of-default (PoD) estimates serve as helpful triggers for adjusting position sizes. Future research might optimize these triggers and rebalancing frequencies, addressing transaction costs and execution complexities.

Developments such as COIN-M futures may further streamline basis trading by obviating stablecoins as collateral.

The evolving ecosystem for crypto trading—encompassing DeFi exchanges, new financial products, and broader institutional involvement—warrants continued research. Attachment to persistent convenience yields and limited arbitrage activity signals deeper structural characteristics of crypto markets. Assessing the impact of factors such as staking returns, slippage, or partial funding constraints will enrich both academic and practitioner insights, enabling more sophisticated and robust long-term portfolio strategies.

Our findings also position crypto convenience yields within the broader carry trade literature. While FX, commodity, and equity futures markets display persistent premia under limits to arbitrage conditions (Lustig, Roussanov & Verdelhan 2011; Du, Tepper & Verdelhan 2018; Gorton, Hayashi & Rouwenhorst 2013), the magnitude of crypto carry—often reaching levels far above those in traditional markets (Schmeling, Schrimpf & Todorov 2023)—underscores the importance of the risk management techniques we develop.

This difference arises because cryptocurrency markets combine unusually high implied leverage, fragmented trading venues, slow settlement processes, and exchange level liquidation mechanisms, which together generate larger and more persistent price dislocations than those documented in traditional asset classes (Makarov & Schoar 2020).

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APPENDIX A. CONFIRMING THE PRESENCE OF CONVENIENCE YIELDS

This appendix provides supporting data on convenience yields in BTC and ETH futures from February 2021 to March 2024. Yield estimates rely on five-minute pricing sourced from Binance quarterly futures, with interpolation of daily USD LIBOR rates for risk-free benchmarks. While proof-of-stake might introduce a crypto-native “risk-free rate,” ongoing concerns about slashing and custody risks justify treating it as zero.

- **A.1 Data and Estimations**

The convenience yield, δ , is inferred through the standard futures pricing equation, where any deviation from cost-of-carry suggests $c > 0$. We annualize yields for comparability but omit the last six trading days before expiry (high variance in annualized returns).

- **A.2 Empirical Findings**

BTC and ETH exhibit average annualized convenience yields of 7.23% and 6.63%, respectively, consistent with Schmeling et al. (2023). Near-dated futures generally deliver higher yields than far-dated contracts (1.76% and 1.77% average spread for BTC and ETH). A regression of weekly and monthly changes in convenience yield against spot price changes shows a positive slope—indicating that yields tend to move in tandem with spot returns.

These findings confirm the ongoing presence of convenience yields in crypto futures, motivating the ARP and arbitrage strategies tested in this paper.

Figures and Graphs

Figure 5: BTC Portfolio opportunity set (as a portion of all possible portfolios)

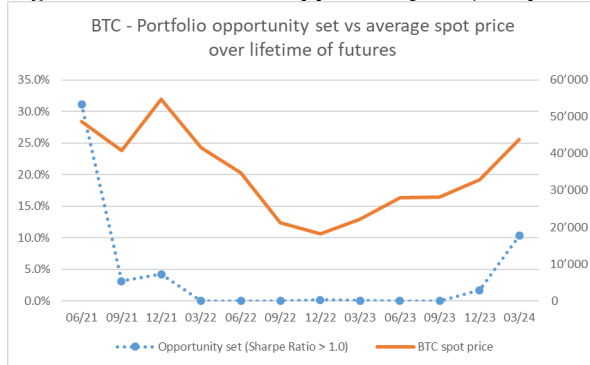


Figure 6: ETH Portfolio opportunity set (as a portion of all possible portfolios)

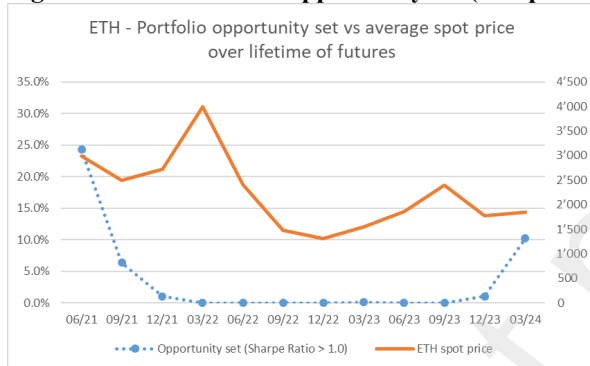


Figure 9 - Sample Rebalancing Event (BTC)

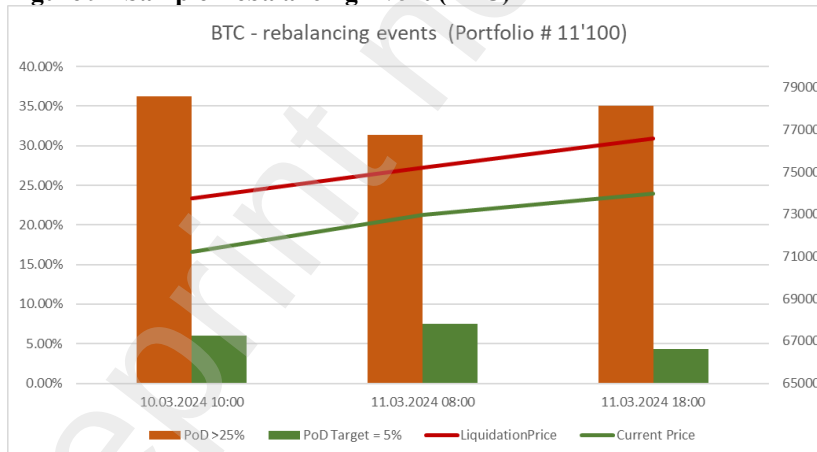


Figure 10 - Sample Rebalancing Event (ETH)

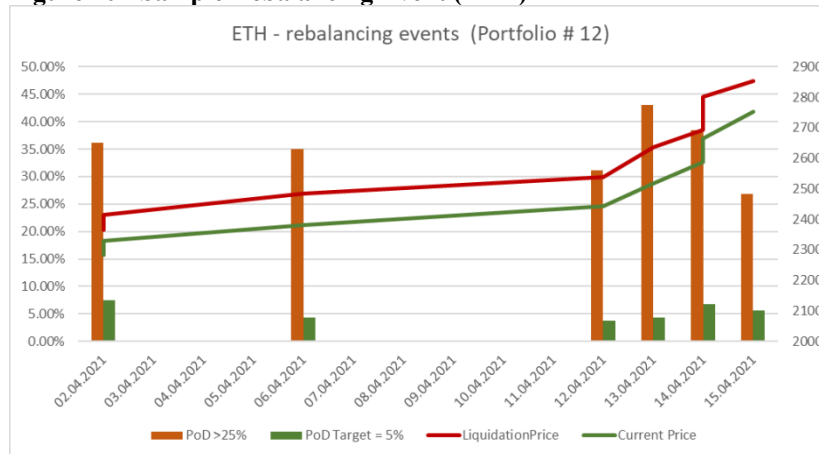


Figure 12: BTC Simulation of prices for one-touch options

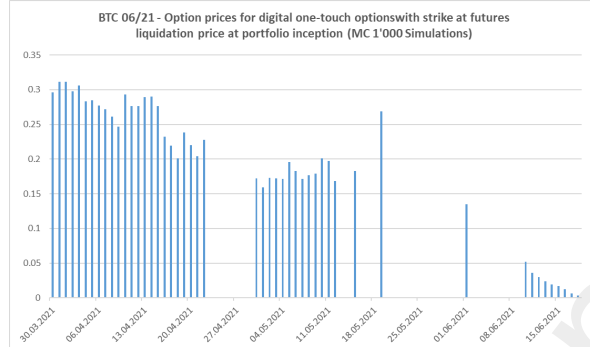


Figure 13: ETH Simulation of prices for one-touch options

